

Discussion 3 Worksheet

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Problem 1: Sum of First n Odd Numbers

Let $P(n)$ be the statement:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

Use the Principle of Mathematical Induction to prove that $P(n)$ is true for all natural numbers n by completing the steps below.

1. **Base Case:** Verify that $P(1)$ is true.

2. **Induction Hypothesis:** Assume that $P(k)$ is true for some $k \geq 1$. Write this assumption explicitly.

3. **Induction Step:** Use the hypothesis to prove that $P(k + 1)$ is true.

4. **Conclusion:** What does this imply about $P(n)$?

Problem 2: Factorial vs Exponential Growth

Prove, using the Principle of Mathematical Induction, that for all integers $n \geq 1$,

$$n! \leq n^n$$

Complete the steps below to structure your proof.

1. **Base Case:** Verify that the inequality holds for $n = 1$.
2. **Induction Hypothesis:** Assume that the inequality holds for some $k \geq 1$. Write this assumption explicitly.
3. **Induction Step:** Use the hypothesis to prove that the inequality holds for $k + 1$.
4. **Conclusion:** What does this imply about the inequality for all $n \geq 1$?

Bonus Exercise: Telescoping Series via Induction

Let the sequence $\{a_n\}$ be defined recursively as:

$$\begin{cases} a_1 = \frac{1}{2}, \\ (n+1)(n+2)a_{n+1} = n^2a_n \quad \text{for } n = 1, 2, 3, \dots \end{cases}$$

Prove, using the Principle of Mathematical Induction, that for all natural numbers n ,

$$\sum_{k=1}^n a_k = \sum_{k=1}^n \frac{1}{k^2} - \frac{n}{n+1} \quad (*)$$

Complete the steps below to structure your proof.

1. **Base Case:** Verify that $(*)$ holds for $n = 1$.

2. **Induction Hypothesis:** Assume that $(*)$ holds for some $k \geq 1$. Write this assumption explicitly.

3. **Induction Step:** Use the hypothesis to prove that $(*)$ holds for $k + 1$.

4. **Conclusion:** What does this imply about the identity $(*)$ for all n ?